



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Smart Courier Tracking and Delivery Route Management Platform

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Review 1 Feedback

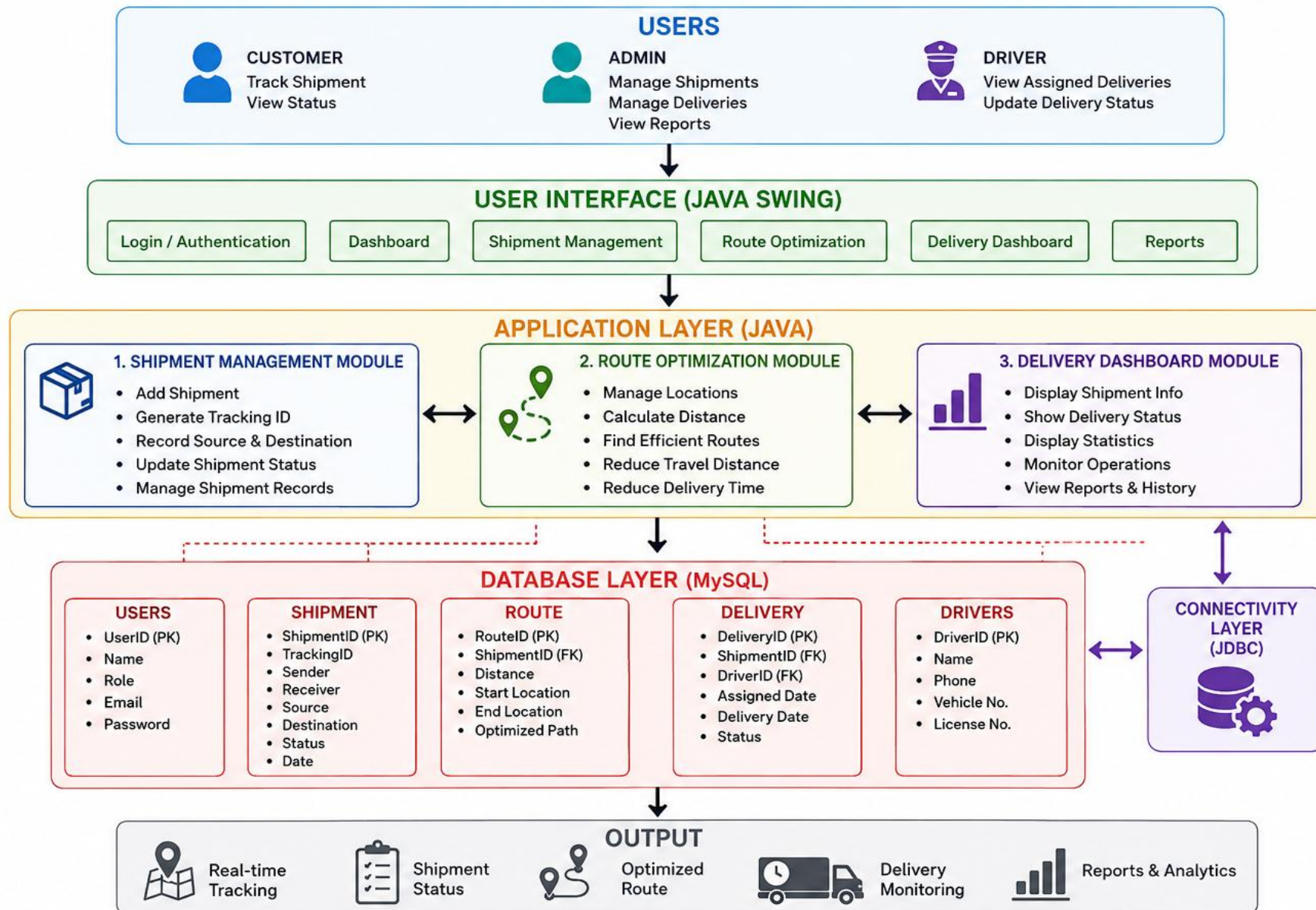
Progress After Review 1

- Refined the overall system architecture.
- Separated the system into **three independent modules**.
- Designed the structure and functionality of the Delivery Dashboard.
- Defined the flow of shipment information through the system.
- Prepared the initial **Use Case and Class Diagram structure**.
- Identified the major database tables and their relationships.
- Defined the Java-based implementation environment.

Current Stage

Detailed System Design → Module Development → Database Integration

Architecture diagram



MODULE 1: Shipment Management Module

Purpose

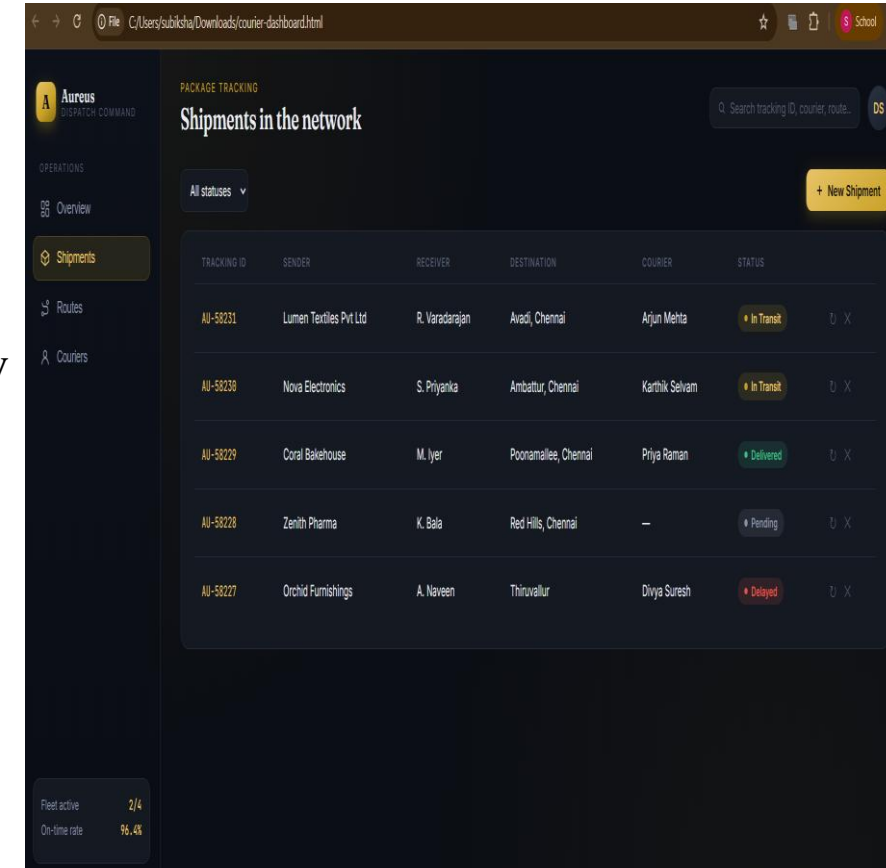
The Shipment Management Module is responsible for creating, storing, updating, and retrieving all shipment-related information throughout the delivery process.

Main Functions

- **Shipment Creation:** Allows the user or administrator to enter new shipment details.
- **Tracking ID Generation:** Generates a unique identification number for every shipment.
- **Customer Details:** Stores sender and receiver information.
- **Location Management:** Records pickup and destination locations.
- **Shipment Status:** Maintains statuses such as Pending, In Transit, and Delivered.
- **Shipment Updates:** Allows authorized users to update shipment information.
- **Record Management:** Retrieves and maintains previous shipment records.

Process

Enter Shipment Details → Generate Tracking ID → Store in Database → Update Shipment Status



Aureus DISPATCH COMMAND

PACKAGE TRACKING

Shipments in the network

Search tracking ID, courier, route. DS

+ New Shipment

TRACKING ID	SENDER	RECEIVER	DESTINATION	COURIER	STATUS
AJ-58231	Lumen Textiles Pvt Ltd	R. Varadarajan	Avadi, Chennai	Arjun Mehta	In Transit
AJ-58238	Nova Electronics	S. Priyanka	Ambattur, Chennai	Karthik Selvam	In Transit
AJ-58229	Coral Bakehouse	M. Iyer	Poonamallee, Chennai	Priya Raman	Delivered
AJ-58228	Zenith Pharma	K. Bala	Red Hills, Chennai	—	Pending
AJ-58227	Orchid Furnishings	A. Naveen	Thiruvallur	Divya Suresh	Delayed

Fleet active 2/4
On-time rate 98.4%

MODULE 2: Route Optimization Module

Purpose

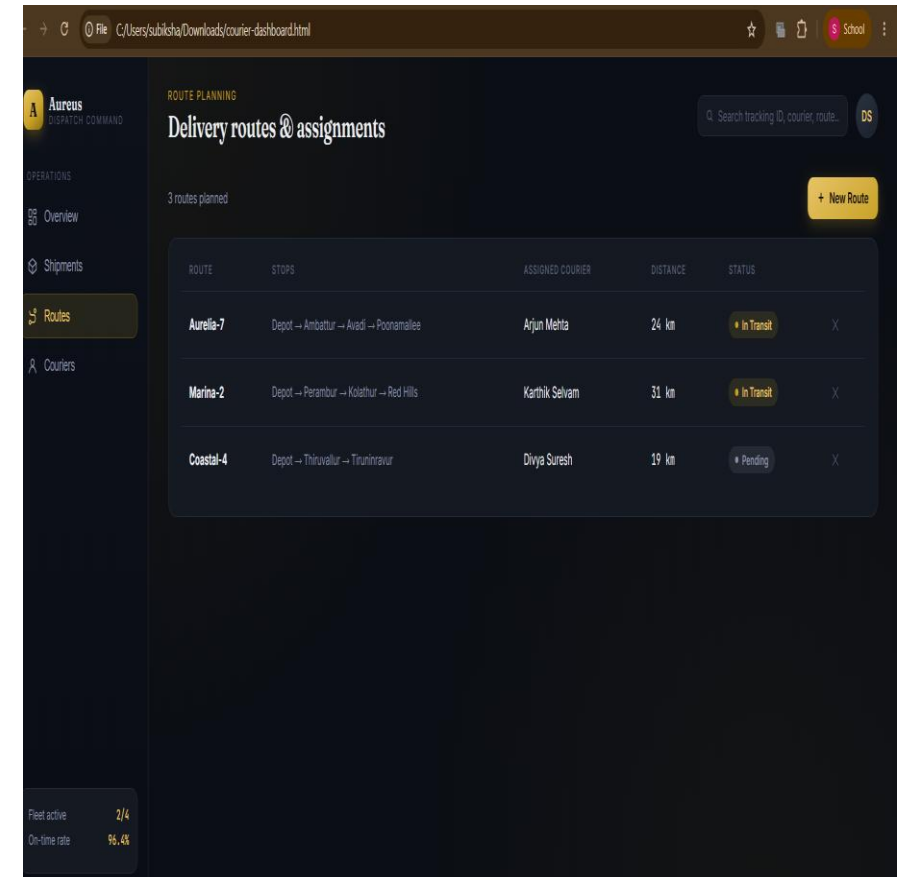
The Route Optimization Module is responsible for identifying an efficient route between the pickup location and destination to improve delivery efficiency.

Main Functions

- **Location Input:** Accepts pickup and destination locations.
- **Distance Calculation:** Determines the distance between delivery points.
- **Route Planning:** Generates possible delivery paths.
- **Efficient Route Selection:** Selects a suitable route based on distance and delivery requirements.
- **Travel Reduction:** Helps reduce unnecessary travel distance.
- **Time Reduction:** Supports faster delivery by selecting efficient routes.
- **Delivery Planning:** Provides route information for delivery assignment.

Process

Pickup Location + Destination → Route Calculation → Route Selection → Delivery Assignment



The screenshot shows a web application interface for route planning. The main heading is 'Delivery routes @ assignments'. Below this, it states '3 routes planned' and provides a '+ New Route' button. A table lists the planned routes with columns for Route, Stops, Assigned Courier, Distance, and Status. The routes are Aurelia-7, Marina-2, and Coastal-4. The status for Aurelia-7 and Marina-2 is 'In Transit', while Coastal-4 is 'Pending'. A sidebar on the left contains navigation links for Overview, Shipments, Routes (selected), and Couriers. At the bottom left, there are summary statistics: 'Fleet active 2/4' and 'On-time rate 96.4%'. A search bar at the top right allows for tracking by ID, courier, or route.

ROUTE	STOPS	ASSIGNED COURIER	DISTANCE	STATUS
Aurelia-7	Depot → Ambattur → Avadi → Poonamallee	Arjun Mehta	24 km	In Transit
Marina-2	Depot → Perambur → Kolathur → Red Hills	Karthik Selvam	31 km	In Transit
Coastal-4	Depot → Thiruvallur → Tirumittur	Divya Suresh	19 km	Pending



Engineer to Excel

MODULE 3: Delivery Dashboard Module



Purpose

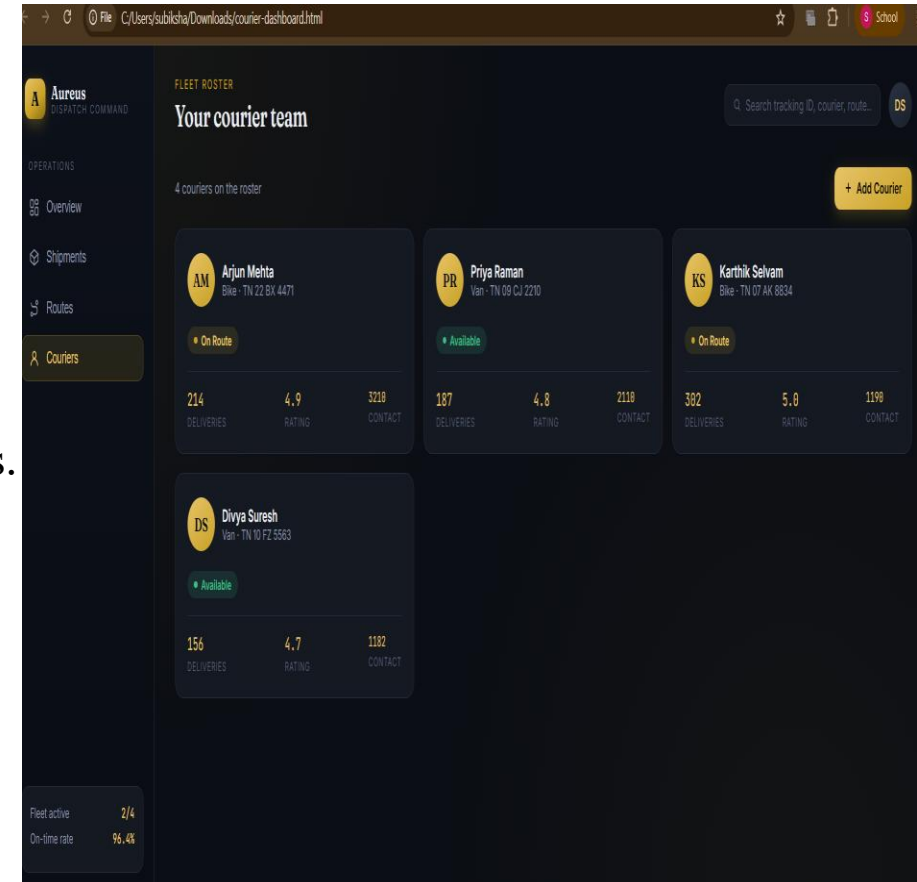
The Delivery Dashboard provides a centralized interface for monitoring shipments, delivery progress, and overall courier operations.

Main Functions

- **Shipment Overview:** Displays the total number of shipments.
- **Status Monitoring:** Shows Pending, In Transit, and Delivered shipments.
- **Delivery Statistics:** Provides summarized information about current deliveries.
- **Shipment Tracking:** Displays the latest status of individual shipments.
- **Operational Monitoring:** Helps administrators monitor delivery activities.
- **History Management:** Provides access to previous shipment and delivery records.
- **Centralized View:** Combines important delivery information in one interface.

Dashboard Indicators

Total Shipments | Pending | In Transit | Delivered



IMPLEMENTATION

Implementation Modules

Login

User enters username and password.
System validates the login.
Successful login opens dashboard.

Shipment

Create shipment.
Generate tracking ID.
Store shipment details.
Update delivery status.

Courier

Add courier.
Store vehicle details.
Assign courier to shipment.

Route

Create route.
Add destinations.
Assign courier.
Store route details.

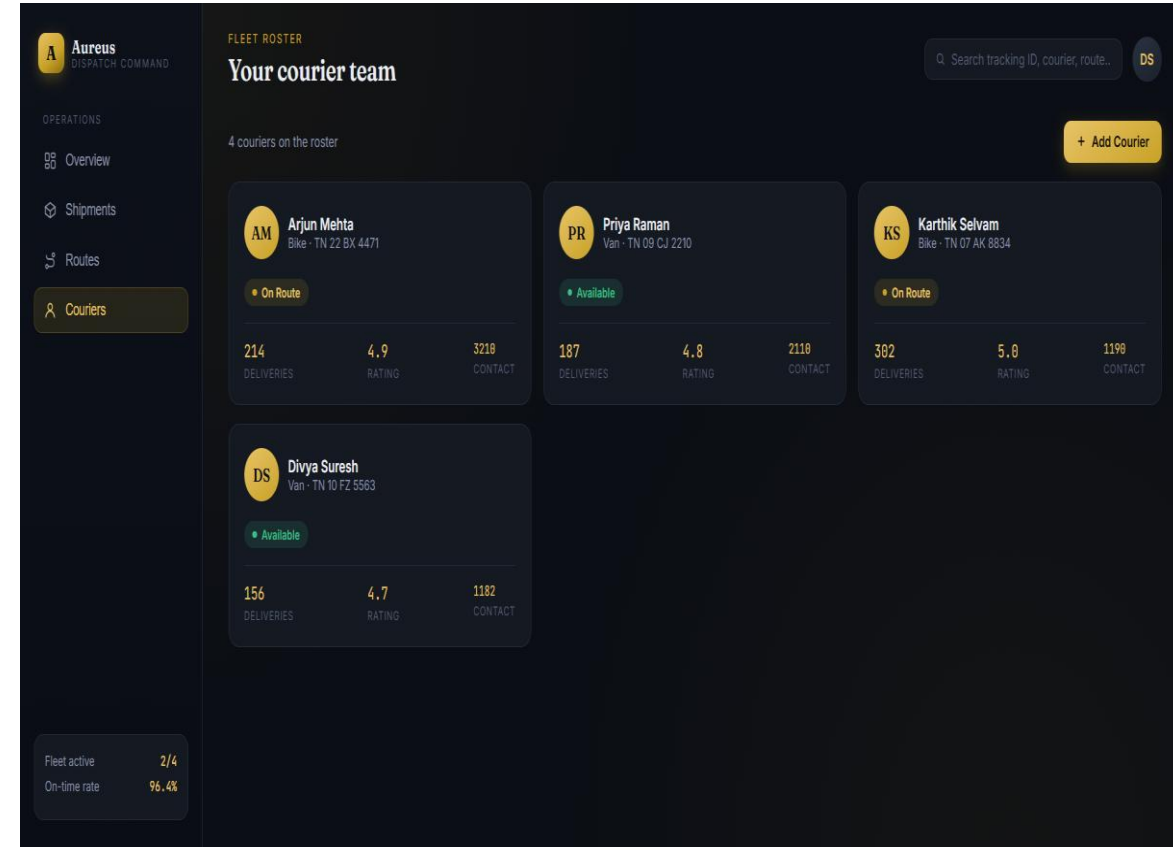
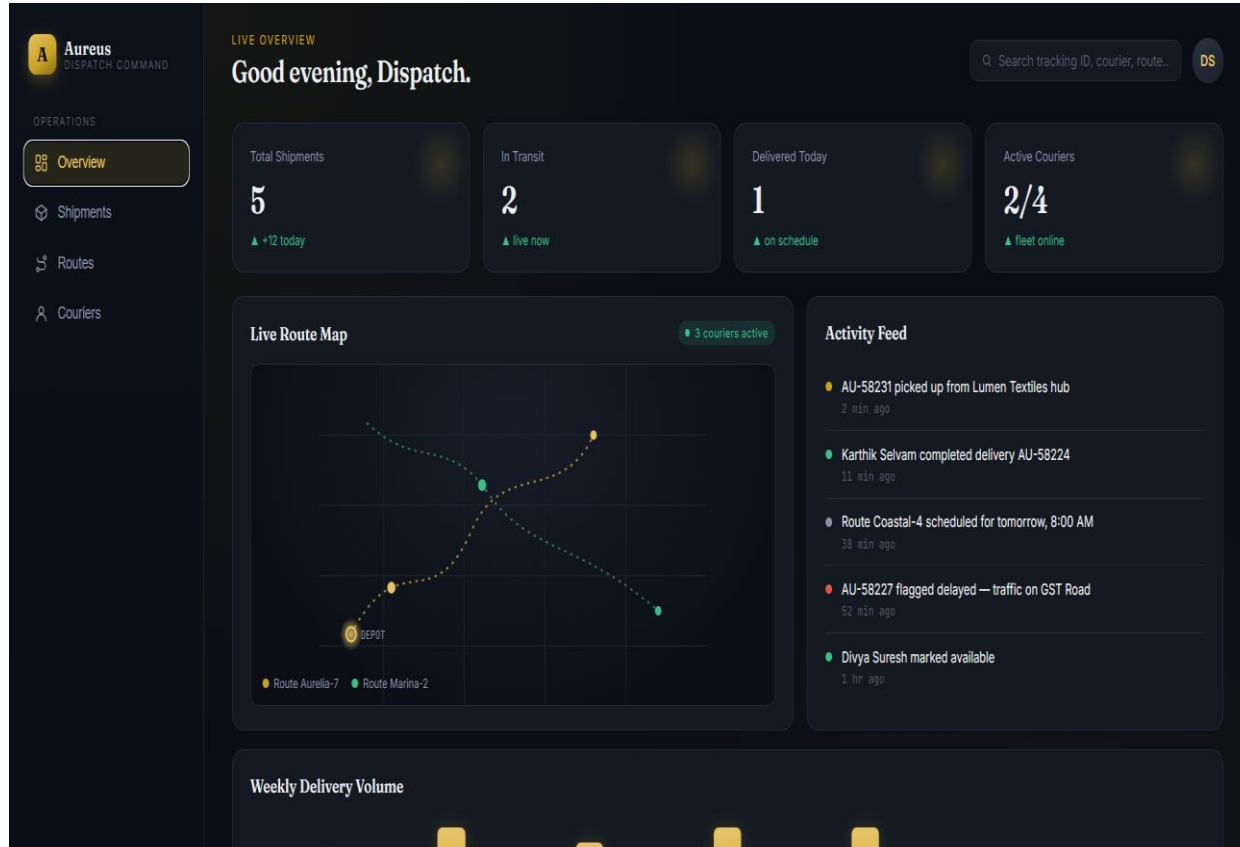
Database

Data can be inserted, updated, deleted
and retrieved.

Completion

**Approximately 60% of the core system
is completed.**

RESULTS



CHALLENGES

Issues

- Managing data flow between the three modules
- Maintaining accurate shipment status
- Handling multiple shipment records efficiently
- Designing a simple and user-friendly dashboard
- Ensuring consistent data between the application and database

Solutions

- Used separate modules with defined responsibilities
- Added structured shipment status management
- Designed organized database tables
- Created a centralized dashboard
- Used JDBC for reliable Java–MySQL communication

THANK YOU!!